

ORIGINAL ARTICLE

The Importance of Context Awareness in PISA Creativity Data

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Received: 6 December 2024 | **Revised:** 27 June 2025 | **Accepted:** 3 September 2025

Keywords: creative thinking | cultural dimensions | educational measurement | PISA | secondary analysis

ABSTRACT

For the first time in 2022, PISA (Programme for International Student Assessment) assessed creative thinking skills among 15-year-old students in 64 countries. The inclusion of measures assessing the generation and evaluation of new and diverse ideas is a welcome step, especially as PISA also includes data on academic performance. Early results from descriptive analysis indicate that girls outperform boys in creative thinking, and better socioeconomic conditions foster improved creative output. This commentary will examine whether these findings from PISA can be generalized to a broader set of countries, relying on cultural differences in creativity as well as education systems. This is to mitigate against the dangers of overgeneralizing or overinterpreting PISA creativity, resulting in the absence of context awareness. Specifically, we will discuss (a) how the characteristics of PISA nations may be systematically different from non-participating nations, along Hofstede's cultural dimensions or cultural distance metrics, (b) how PISA results can be merged with other existing country-level datasets to enrich interpretations, (c) implications of broad PISA results against the background of national-level metrics like gender gap or human development indices, and (d) future directions for valuable large-scale datasets such as PISA.

Starting in 2000, the Programme for International Student Assessment (PISA) has assessed educational achievement using items representing real-world challenges in the domains of science, mathematics, and reading (Sjøberg and Jenkins 2022). PISA is administered to individuals aged 15 years across OECD (Organization for Economic Cooperation and Development) countries every 3 years; in the most recent edition in 2022, creative thinking skills were assessed for the first time in a subset of 64 PISA nations (Grey and Morris 2024; OECD 2024). This inclusion provides numerous opportunities to examine the linkage between creativity and academic performance (see also Gajda et al. 2017) as well as comparisons within creative thinking across nations in youth samples. Further, PISA creativity data allows researchers to evaluate education systems and

policies, providing impetus to teaching for creativity (Jeffrey and Craft 2004).

This commentary aims to persuade future researchers using the PISA creativity dataset to apply context awareness when exploring relationships within and between nations and PISA constructs. Borrowing from computer science, we introduce context awareness (Hong et al. 2009; Seaver 2015) as a concept that can be applied to large datasets, such as PISA. Next, we argue that PISA nations may systematically differ from non-PISA countries on multiple dimensions, limiting the generalizability of inferences across outcomes. Context awareness can also facilitate a meaningful combination of PISA data with other existing international datasets to provide richer contextualization

of results. We will provide one illustrative example of applying national-level metrics to enrich interpretations using the PISA finding that girls outperform boys on all creative thinking measures (OECD 2024). Last, we will outline the next steps and future directions for making the most out of large datasets on creative thinking, like PISA.

1 | Context Awareness

In the early 1990s, when computing systems were just taking shape, researchers introduced context awareness as “the ability of a mobile user’s applications to discover and react to changes in the environment they are situated in,” (Schilit and Theimer 1994, 3). Dey and Abowd (2000) define context as “any information that can be used to characterize the situation of an entity (3).” An entity could be a person, place, or object (and for the purposes of this paper, a dataset or additional variables) that is relevant for the interaction (or use) under examination. Today, you may recognize it as your phone’s location tracking you and, therefore, suggesting the best places to eat in the neighborhood, based on your past search history on a food delivery app. Context awareness goes a step further when integrated with big data analytics¹ to provide the optimal user experience (Dinh et al. 2020). This can be based on your history, preferences, and even social location,² ranging from curated music playlists to friend or follower suggestions on social media platforms. These are typically referred to as recommender systems, which are meant to use multiple criteria to decide exactly what information to use, and how to use it (Zheng et al. 2019). Today, contextually aware systems are the bedrock of modern computing, continually updating themselves based on new knowledge about the environment.

Notably, context-awareness in computing systems is just one framework to understand how context matters in using research data or evidence.³ There is extensive work in other domains that already discusses the role of, say, organizational or political context, that could affect how data is used (Coburn and Turner 2011), such as the time available to process data, indicators on which data are available, existing power relations, among others. Education research also suggests that where the data are produced and who produces the data also affect stakeholder decisions (Ion et al. 2019; Langrall et al. 2011). To a large extent, these overlap with the ideas embedded within context-awareness in computing systems (e.g., time, place/location, etc.).

Applying this concept to large-scale international surveys emphasizes the importance of integrating context awareness when we interpret static data, especially aggregate data (e.g., one that is meant to represent an entire country, as a singular unit of analysis). In the absence of contextualizing results, there may be errors in overgeneralizations or misinterpretations of valuable data. The classic example here is an age-old tale of spurious correlation that statisticians typically use: how do we understand the reasons underlying high homicide rates? The example goes on to show that if you plot ice cream sales and homicide rates side by side, they show a high degree of correlation. Without context awareness, it would be easy to conclude that ice cream sales and homicide rates are somehow linked, when in fact there are common factors like warm temperatures that could explain

both phenomena (Salkind 2013). Thus far, applications of context awareness in research have been largely restricted to studying big data (Dinh et al. 2020), which deals with how to leverage the available datasets to better inform the criteria for developing context-aware systems. However, there may be novelty and value in applying context awareness to large-scale datasets that deal with policy-relevant outcomes. For example, Bibri (2018) outlines the importance of context-aware computing using data from smart cities to enhance our understanding of urban sustainability. It is therefore possible that context awareness has broader value for insights beyond the domain of computing.⁴

For a more relevant example, consider a PISA finding such as large gaps in creative thinking across nations and economies (OECD 2024). This can be contextualized by understanding differences in say, GDP or other socioeconomic indicators at the country level. With respect to PISA data, context awareness may be applied more easily when looking at relevant data from a single nation or similar ones. The challenge for informed statisticians and researchers is, therefore, to use context to define what constitutes *relevant* data. For instance, given that the authors of this article reside in India, we were curious to understand why the most populous nation in the world did not participate in PISA. This led us to several articles and commentaries about how India ranked near the bottom of the 2009 PISA round, followed by the Government of India boycotting the test due to unfair and unfamiliar language, then agreeing to participate in the 2022 round if the OECD contextualized items for Indian students, but withdrawing due to COVID-19’s impact on education systems (Addey and Gorur 2020; Lab Team 2022; Rajagopalan 2019). That’s a fair bit of context already, and we have not even discussed the sociopolitical milieu of India with respect to education policy and standardized testing.

As the PISA creativity assessment is the core focus of this paper, there are other PISA domains (reading, mathematics, and science) that could also benefit from a careful application of context. In fact, work by Perry and Ercikan (2015) recommends revisiting the rankings provided by PISA scores since they may not fully take into account socio-economic status and home environment factors that vary widely across countries. As Liang (2010) shows, having rich contextual data (for the US) can help better situate mathematics scores, although it is not clear if it is possible to collect this same data for all countries. Schmidt and Burroughs (2016) survey PISA math scores across countries and argue that interpreting inequalities therein across countries requires a nuanced understanding of education policy contexts. For the PISA science scores, Oliver et al. (2021) talk about the importance of inquiry-based science education and tailoring these to specific countries. Finally, Tan (2020) shows the importance of understanding PISA reading scores in the context of reading habits and cultural capital (which considers students’ socioeconomic backgrounds, as well as their tastes and preferences). This suggests that there is already a wide strand of literature that examines the interplay between context and PISA scores, albeit not using context-awareness, and not using PISA creativity scores, which is the focus of this paper.

Similarly, it is important to apply context awareness when comparing nations on indices such as PISA creativity. This allows broadening the focus on the characteristics of participating

nations, such as the demographic distribution of the country, as well as incorporating these macro-level indicators to strengthen insights from PISA creativity results. It is widely acknowledged that contextualizing the PISA test items across several nations and languages is already problematic (Addey and Gorur 2020; Sjøberg and Jenkins 2022, 7). In a similar vein, we argue that it is equally problematic to interpret PISA results in the absence of larger contexts.

2 | Features of PISA Creativity Nations Compared to Missing Countries

Although 81 countries participated in PISA 2022, data from a subset of 64 nations are available for PISA creativity analyses (OECD 2023, 2024). This adds an additional layer of missingness to the dataset along with the nations that did not participate in PISA at all. It is likely that some countries did not participate in the PISA 2022 cycle due to varying political, economic, or logistical issues. Although the list of excluded countries is extensive, examples include several low- and middle-income nations in Africa, the Middle East, and South Asia. To understand the differences between PISA and non-PISA nations, we ran a quick and rough estimate on a few national-level indicators from the World Bank Data Bank (World Bank 2024).

The results are displayed in Table 1. First, there is a statistically significant difference, on average, between PISA and non-PISA participating countries on GDP per capita. On average, countries participating in the PISA have nearly double the per capita incomes of non-participating countries. Similar differences are observed on various socio-demographic indicators, including poverty (non-PISA countries are significantly poorer), population growth (PISA countries have lower growth rates), and structure of the population (non-PISA countries are predominantly rural and have higher dependency ratios). Even in variables that capture educational system characteristics, we see significant differences between PISA and non-PISA countries. For example, on average, 11% of the population aged 25 years or older in non-PISA countries have a Bachelor's degree, compared to nearly 23.7% in PISA countries. However, we find no statistically significant differences between PISA and non-PISA countries in the share of education in GDP as well as government expenditures. This indicates that, on average, the level of investment in education may not differ across PISA and non-PISA countries, but there may still be other systemic differences in educational inputs.

Last, there are no statistically significant differences in employment indicators between PISA and non-PISA countries. There is a larger share of the population with basic education in non-PISA countries (nearly 50% on average), but this difference is not statistically significant after correcting for multiple hypothesis testing. On average, there is a substantial difference in youth not in employment, education, or training in non-PISA countries (approximately 23.6%) compared to PISA countries (14.3%). This suggests that there could be structural differences in youth between PISA and non-PISA countries.

In sum, PISA countries differ systematically from non-PISA countries on a range of socioeconomic and demographic

characteristics. Research has already suggested a slight positive link between creativity and socioeconomic status (Acar et al. 2023), making it important to factor this into any conclusions drawn about creative performance. Crucially, there are also statistically significant differences in education-related characteristics, suggesting that education systems differ markedly between PISA and non-PISA countries. There are also large and statistically significant differences between PISA and non-PISA countries on indicators of innovation. The latter file far more patent and trademark applications (although this is largely driven by the United States and China as non-PISA countries). Interestingly, the PISA countries on average scored 25% higher on the Global Innovation Index (GII) between 2011 and 2024 and on the Global Creativity Index (GCI) score (Florida and Mellander 2023). This preliminary analysis suggests that making broad generalizations about creativity in the area of education based on data from PISA countries alone may lack context and therefore present an incomplete picture.⁵

3 | Merging PISA Across Datasets

This is not meant to be a methodological commentary, and therefore we will not discuss the best practices regarding how cross-country or national systems datasets should be merged (Castellacci and Natera 2011; Liu et al. 2024). However, we present a few instances of existing and recent datasets that can be integrated with PISA to yield more holistic insights:

3.1 | Cultural Distance

To address the WEIRD (Western, Educated, Industrialized, Rich, and Democratic) problem in psychological studies, researchers developed measures of cultural and psychological distance using data from the World Values Survey to understand how far or close societies are from another nation, like the US (Muthukrishna et al. 2020). Cultural distance can help set the context against which comparisons are made between countries, while also recognizing that within-nation variation exists. In terms of PISA, incorporating this dataset can provide a nuanced understanding of PISA differences, not only in creative thinking skills but in other core areas of academic performance as well.

3.2 | World Values Survey

The World Values Survey (WVS) is a long-running survey by social scientists that studies the beliefs, motivations, and values of individuals globally (Inglehart et al. 2014). The most recently available version of the dataset is the WVS 7, which was released in 2022 (Haerpfer et al. 2022). It has a specific component related to data from Europe, which is called the European Values Survey (EVS). Among several other dimensions, WVS contains useful information on what individual beliefs around important qualities in children are (e.g., independence, imagination, obedience, etc.), attitudes toward education, and institutional trust. These may help better situate a country's performance on the PISA based on collective (or aggregate) beliefs around children and educational systems from the WVS.

TABLE 1 | Summary statistics and test for differences.

	Non-PISA country	PISA country	Overall	Test (<i>p</i>)
GDP per capita growth	1.086 (3.349)	1.895 (1.290)	1.325 (2.917)	0.067
GDP per capita (PPP, constant 2021 international \$)	19,471.170 (23,750.304)	39,807.004 (24,484.416)	25,838.956 (25,721.101)	< 0.001
<i>Socio-demographic factors</i>				
Multidimensional poverty headcount ratio (UNDP) (% of population)	28.479 (26.725)	2.887 (2.587)	22.834 (25.885)	< 0.001
Population ages 0–14 (% of total population)	29.495 (10.422)	19.109 (5.688)	26.527 (10.426)	< 0.001
Population growth (annual %)	1.230 (1.188)	0.532 (1.011)	1.030 (1.181)	< 0.001
Poverty headcount ratio at \$2.15 a day (2017 PPP) (% of population)	14.964 (20.678)	0.923 (1.351)	9.150 (17.266)	< 0.001
Age dependency ratio (% of working-age population)	61.428 (17.377)	49.820 (8.712)	58.111 (16.259)	< 0.001
Rural population (% of total population)	44.038 (24.598)	25.098 (15.089)	38.577 (23.846)	< 0.001
<i>Education system</i>				
Government expenditure on education, total (% of GDP)	4.396 (2.201)	4.597 (1.166)	4.461 (1.929)	0.512
Government expenditure on education, total (% of government expenditure)	14.025 (5.092)	13.692 (4.299)	13.923 (4.854)	0.667
Educational attainment, at least Bachelor's or equivalent, population 25+, total (%)	11.261 (10.097)	23.735 (8.502)	15.347 (11.235)	< 0.001
Primary education, duration (years)	5.884 (0.773)	5.472 (0.873)	5.763 (0.823)	< 0.001
Secondary education, duration (years)	6.308 (0.824)	6.613 (1.006)	6.398 (0.890)	0.023
Trained teachers in lower secondary education (% of total teachers)	78.416 (21.209)	90.179 (19.058)	82.076 (21.177)	0.014
<i>Employment</i>				
Employment to population ratio, ages 15–24, total (%) (modeled ILO estimate)	34.408 (14.169)	34.962 (12.716)	34.592 (13.672)	0.795
Labor force participation rate, female (% of female population ages 15+) (modeled ILO estimate)	49.564 (16.354)	52.584 (9.887)	50.565 (14.574)	0.183
Labor force with basic education (% of total working-age population with basic education)	50.680 (14.864)	45.021 (16.176)	48.644 (15.540)	0.025

(Continues)

TABLE 1 | (Continued)

	Non-PISA country	PISA country	Overall	Test (p)
Share of youth not in education, employment or training, total (% of youth population)	23.644 (9.966)	14.294 (6.962)	20.544 (10.079)	< 0.001
<i>Innovation and creativity indicators</i>				
Patent applications by residents	28,634.803 (165,444.489)	4641.684 (22,411.109)	17,263.341 (121,545.765)	< 0.001
Trademark applications by residents	158,384.594 (939,454.386)	35,668.739 (61,941.324)	101,210.960 (690,420.751)	< 0.001
Average score on Global Innovation Index (GII, 2011–2024)	30.621 (12.274)	41.137 (10.069)	35.394 (12.476)	< 0.001
Score on Global Creativity Index	0.419 (0.220)	0.655 (0.202)	0.518 (0.242)	< 0.001
N	155 (69.8%)	64 (30.2%)	219 (100.0%)	

Note: Averages reported for each indicator between 2017 and 2023 for all available countries. Classification of PISA and non-PISA countries based on participants in 2022 PISA surveys. The test column reports *p*-values from a single linear regression of the dummy variable (PISA) on the variable. $p < 0.0027$ (Bonferroni-corrected *p*-value).

3.3 | World Bank Indicators

Of the numerous determinants of creative thinking performance recorded in PISA, country-level socioeconomic factors could play a significant role in explaining differences. Literature has previously shown that socioeconomic status (SES) matters for performance on the PISA in a nuanced way (Acar et al. 2023; Gamazo and Martínez-Abad 2020). For example, country-level education expenditures (typically measured as a share of the Gross Domestic Product, or GDP) could help researchers better understand the role that educational inputs play in PISA achievements. Overall levels of economic development as captured by macroeconomic indicators (like the GDP per capita) should be considered when it comes to comparing education outcomes across countries. This moves beyond the debates around capturing SES *within* the PISA survey itself (Avvisati 2020), but rather accounts for whether there are systematic differences in the kind of countries that participate in PISA. This type of selection bias could obfuscate a broader interpretation of the PISA results, limiting generalizability to strictly OECD countries (which are typically among the wealthiest countries in the world, J. Lee and Borgonovi 2022).

3.4 | Human Development Index

The Human Development Index (HDI) was formulated by the United Nations Development Programme (UNDP) as a holistic approach to measuring development globally (Anand and Sen 1994). Although various extensions to the HDI have been proposed (Ranis et al. 2006), at its core, the measure offers insights into factors related to human capital that are not captured by typical aggregate economic indicators. For instance, aspects of human capabilities (e.g., health status or other educational inputs) that are accounted for in the HDI could explain differences in PISA scores between countries. In fact, some of the more modern measures of human development, such as the Human Capital Index (Kraay 2019), use PISA data to better understand

how human capital accumulation occurs in the areas of health and education.

3.5 | Global Innovation Index (GII)

This index, currently in its 17th edition, benchmarks nations and economies based on a ratio of innovation input (e.g., gross expenditure on research and development or number of graduates in science and technology) and output (e.g., number of national feature films produced or the percentage of creative goods exports) indices, arriving at an overall country ranking (Dutta et al. 2024). The GII provides a macro-level estimate of the value of innovation in a country, which trickles down to creative expression at an individual level. Therefore, merging GII with PISA data also helps in drawing a link between the scale of creativity from a personal to national level.

3.6 | Global Creativity Index (GCI)

The GCI examines variations in national creativity ecosystems across talent (educational attainment and the share of the workforce employed in creative jobs), tolerance (freedom of expression and tolerance for gays and lesbians), and technology (innovation, and research and development) in 160 nations (Florida and Mellander 2023). Similar to the GII, GCI rankings can be used to contextualize the salience of creativity within a country from a human and social capital perspective.

3.7 | Global Gender Gap Index

This index annually ranks countries and economies on gender parity across four dimensions: economic participation and opportunity, educational attainment, health and survival, and political empowerment (World Economic Forum 2024). For PISA data, the subindex on education may be the most relevant; however, it may be useful to consider the gender gap index holistically to examine

how 15-year-old girls' creative thinking skills extend to real-world contexts across nations. Situating PISA creativity results within this broader context is more likely to help researchers consider the long-term implications of these results.

4 | Making PISA Creativity Results More Contextually Aware: An Example

...in nearly all countries and economies, girls performed equally to or outperformed boys, on average, in tasks in all domains and in all ideation processes.

(OECD 2024, 142)

The creativity assessment included tasks that encouraged expression (written or verbal) and problem-solving (social or scientific). The result that girls outperformed boys across the board is potentially salient, having wide-reaching implications for fostering creativity in education for all genders. Research shows that there are seemingly fewer discrepancies between men and women when it comes to creative potential, but larger differences with men outperforming women in creative achievement (Taylor 2023). In the PISA context, there exist gender differences in effortful responding on tests, domain- and task-specific creative performance, as well as developmental dynamics at play within this age cohort, that is, 15 years and 3 months to 16 years and 2 months. Therefore, caution must be exercised when interpreting gender differences based on the overall PISA creative thinking score (Taylor 2025). These are significant methodological and interpretative issues that are beyond the scope of this commentary, but merit careful consideration.

However, given the descriptive exercise of this paper in understanding the role of context, we use the currently available overall PISA creativity score to explore: Do girls have an advantage when entering Science, Technology, Engineering, and Math (STEM) careers that involve scientific creativity? Or could boys be less innovative in the real world? How can we further contextualize this finding?

First, consider whether gender differences in creative thinking (as measured by the PISA test) may be linked to gender gaps in what one chooses to study at the tertiary (or undergraduate) level. Figure 1 shows that there is a moderate and positive correlation ($r = 0.39$, $p = 0.002$) between females choosing to study Arts or Humanities (UNESCO Institute for Statistics 2020) and the difference in creative thinking test scores in PISA. Notably, a similar trend is observed in the case of females in STEM fields at the undergraduate level, but the correlation is not statistically significant. If we consider enrollment in educational fields as a criterion related to PISA creativity, perhaps the PISA items capture expression-based creativity (arts and humanities) better as compared to scientific creativity (STEM). These estimates are based on cross-sectional data and not time-series data; therefore, we cannot conclude whether a girl scoring high on PISA creativity will or will not go on to pursue arts/humanities or STEM courses. However, the trends suggest that PISA creativity items may be more strongly correlated with pursuing education in the arts as compared to STEM fields.

Aside from this, it is possible that the correlation reflects broader factors that link the creativity gender gap with the share of females pursuing arts and humanities. For instance, York (2008) finds that among gifted high school valedictorians,

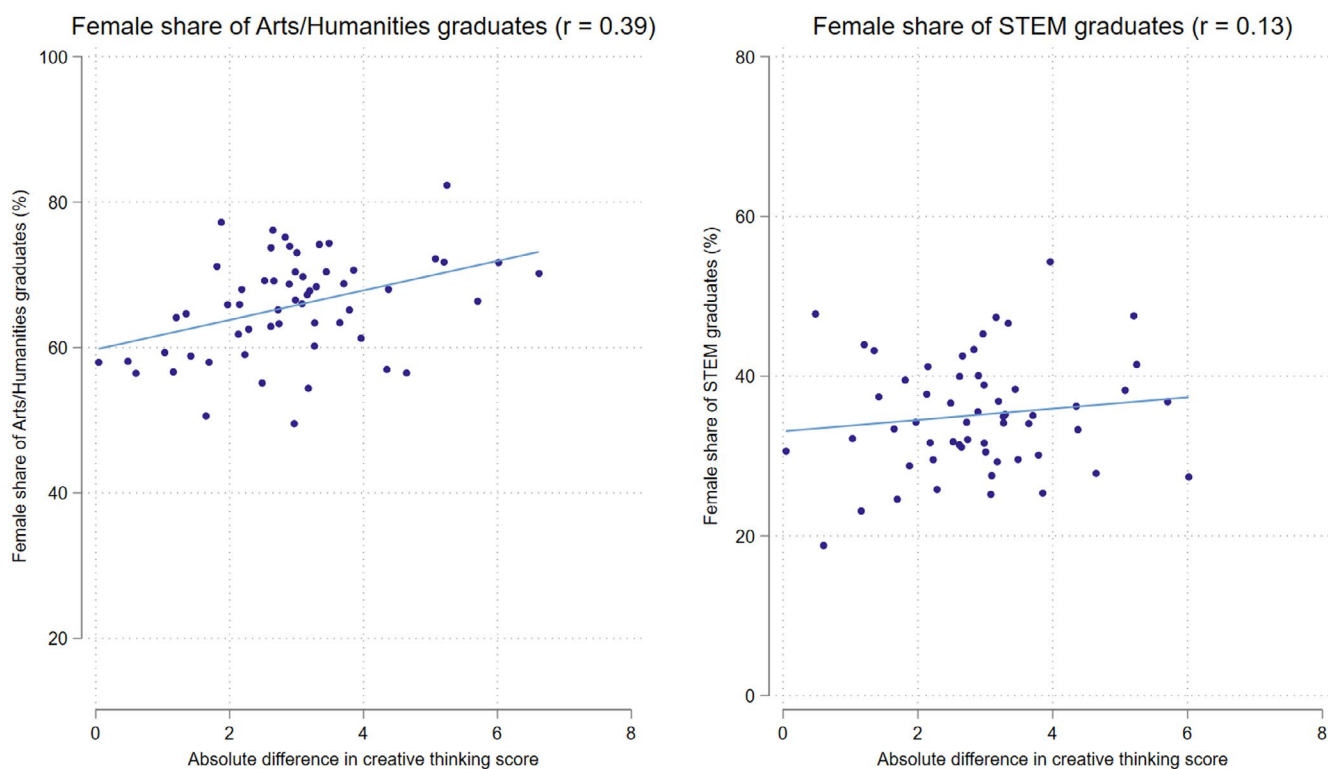


FIGURE 1 | Correlation between gender difference in PISA creative thinking score and gender gaps in field of study.

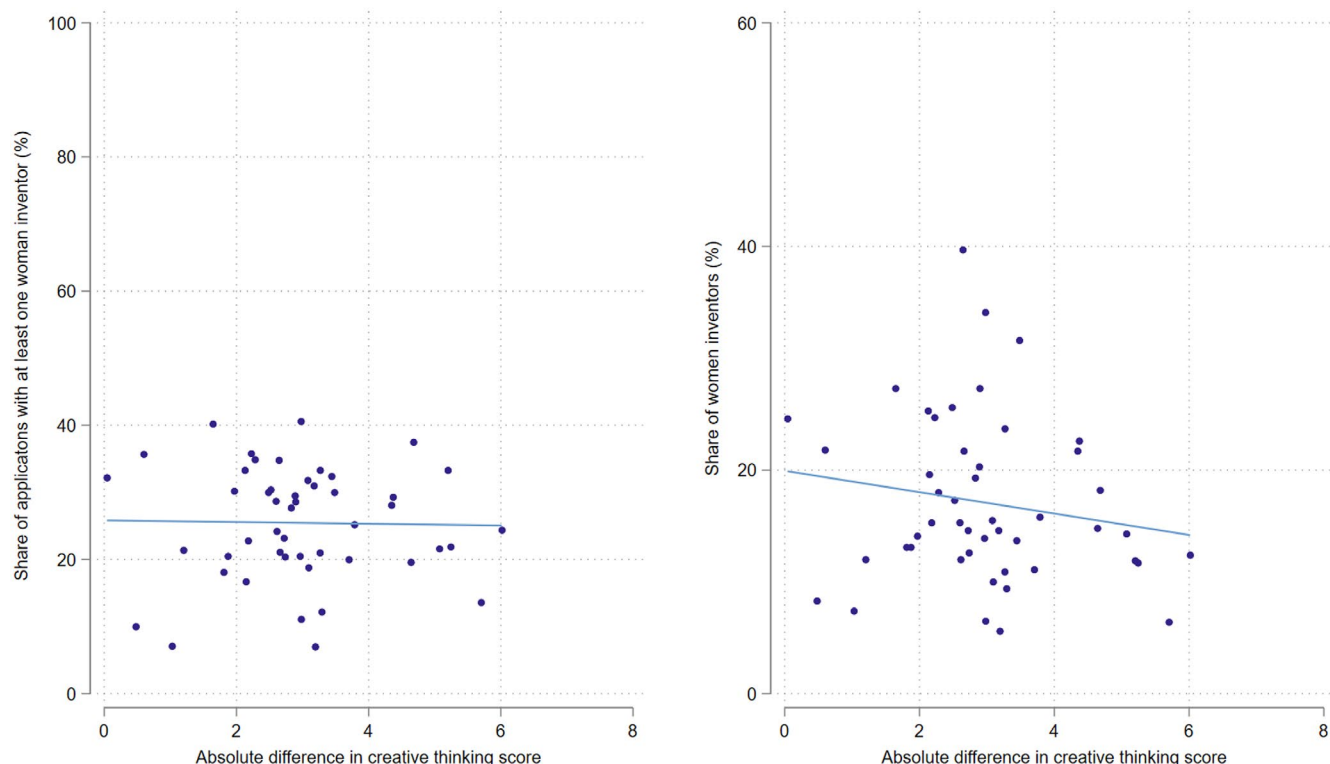


FIGURE 2 | Correlation between gender difference in PISA creative thinking score and gender gaps in innovation.

males are more likely to choose STEM majors as compared to females who are more likely to choose arts/humanities majors; this study was done using data from North Carolina, USA. For high-ability first-generation college females in the USA, the choice of a (arts/humanities) college major was related to identity exploration (Speirs Neumeister and Rinker 2006). Moreover, although research has examined differences in creativity across college majors for gifted students (e.g., Miller and Smith 2017) as well as gender differences in creativity across nations (e.g., Nakano et al. 2021), the role of gender differences in creativity associated with college major choice cross-culturally is yet to be examined. In this case, additional cross-cultural context does not exist yet, which in itself provides fodder for future research.

The PISA gender gap in creativity raises another interesting question: if girls outperform boys in creative thinking in adolescence, does this manifest in any broader indicator of creative achievement? To explore this, Figure 2 plots the gender gap in PISA creativity scores with two indicators of the gender gap in innovation at the country level from the World Intellectual Property Indicators (World Intellectual Property Organization 2025). The figure suggests that there is either no correlation ($r = -0.02$, $p = 0.9$) or a weakly negative and non-significant one ($r = -0.17$, $p = 0.23$). This means that girls outperforming boys in PISA creative thinking in a particular country does not relate to the share of women inventors or the share of women on patent applications. This suggests that the PISA creativity gender gap may not necessarily translate into downstream outcomes related to innovation and creativity.

For broader contextualization, consider how the gender gap in the PISA creativity score lines up with country-level measures

of gender gaps across different dimensions (World Economic Forum 2024). A higher score on the Gender Gap Index means more parity (or equality), and a lower score means less parity between genders in that country. Figure 3,b captures the correlation between the gender differences in PISA creativity test scores and the overall Global Gender Gap Index (Figure 2a) and the subindices (Figure 2b). At the aggregate gender gap index, we find no significant correlation between gender differences on the PISA creativity test and the overall Global Gender Gap Index ($r = -0.04$, $p = 0.74$). This means that a gender gap in creativity does not necessarily imply an overall gender gap in these PISA countries. The correlations from the GGI subindices tell a more nuanced story: there is a statistically significant and negative correlation between the health GGI and gender difference in creativity scores ($r = -0.36$, $p = 0.003$), implying that a larger gender gap in creativity was linked with a larger gender gap in health and survival indicators in the PISA countries. The subindex includes gender differences in the sex ratio at birth and life expectancy, suggesting that in countries where girls score more than boys on creative thinking, there could also be structural-demographic issues related to the “missing women” problem.

5 | How Much Context Is Enough Context?

We acknowledge that when a dataset like PISA creativity is released, it presents a novel opportunity for researchers to draw inferences and make policy recommendations, more so because creativity is touted as a 21st-century skill (Grey and Morris 2024). However, academics may tend to parcel out the data for one country or region and focus on the implications of results for those nations alone. Instead, when researchers aim to compare numerous (if not all) nations, we urge them to apply

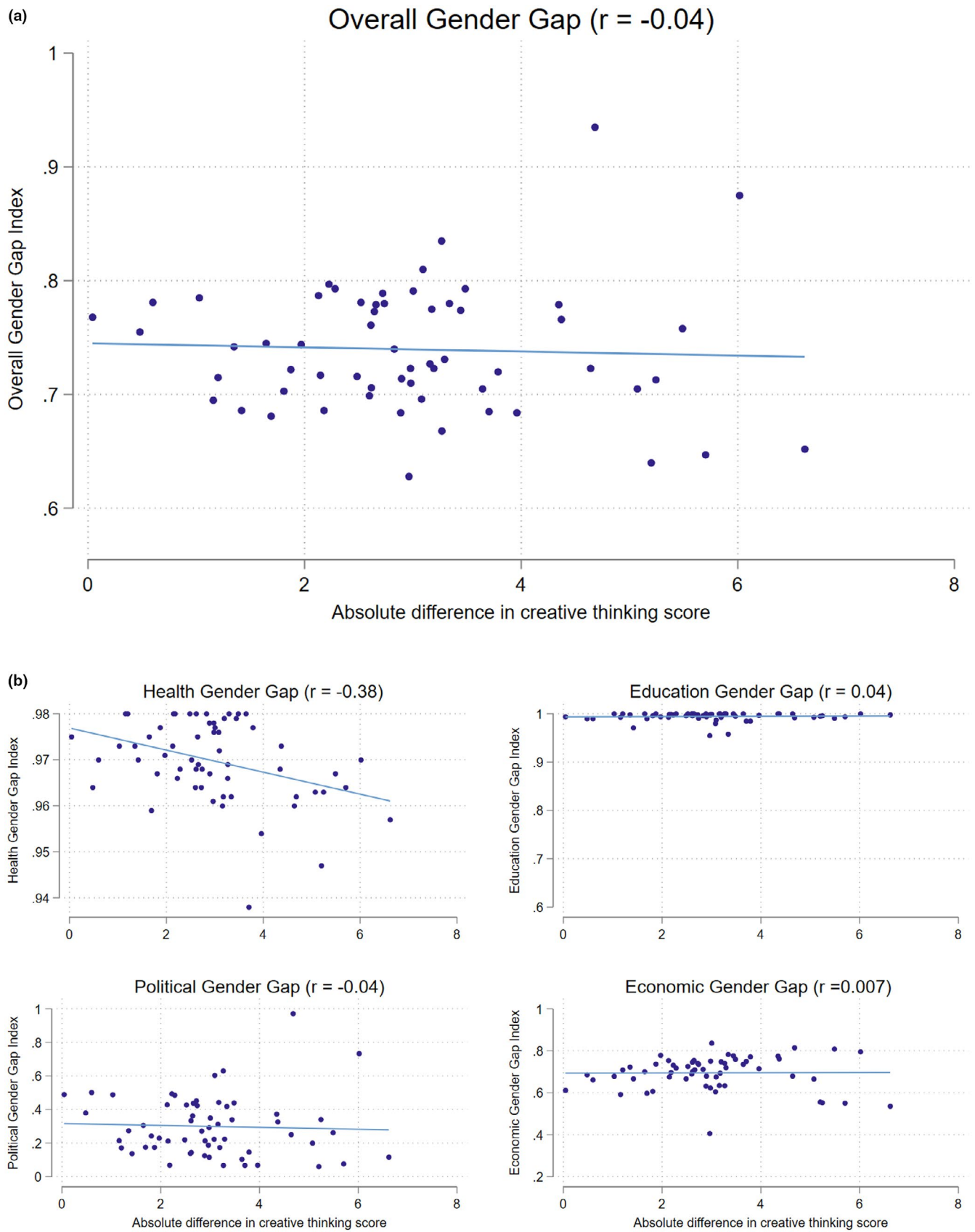


FIGURE 3 | (a) Overall gender gap index and gender difference in PISA creativity test score. (b) Gender gap subindices and gender difference in PISA creativity test score.

context awareness in their analyses. Through this commentary, we have aimed to provide some examples of country-level estimates and contextual data that can be merged with PISA creativity; however, it is vital to state that the comparisons or mergers with other databases that are proposed in this commentary are at the country level and not at the individual level. We are also very cautious about not making causal inferences, as merged data across indices will nearly always yield associations and broad trends.

It is also important to acknowledge that we do not deal with how context is applied when using or assessing insights from qualitative data. This may be particularly relevant in education, where policymakers may overweight quantitative data from exercises like PISA, over context-rich information from qualitative studies (Wiseman 2010). There are a few comparative qualitative studies in education that help better understand context and how to use it to infer cultural or societal differences (Lee 2016; Lee and Yang 2017). One avenue for future work is to determine a framework for a mixed-methods approach to understanding and integrating context into comparative studies of education or education policy.

So, when should we stop expanding the context around a data-frame? To adapt a quote attributed to Karl Popper, although we can omit several contextual variables when analyzing static multi-country data like PISA, the oversimplification needs to be systematic.⁶ This commentary does not propose to include all potential variables that could plausibly be connected to PISA creativity in every analysis; however, we argue that it is important to include a few directly relevant country-level indicators to better situate PISA creativity findings. PISA, just like several of the other datasets discussed in this paper, is not classified as Big Data; this ever-changing data is characterized by volume, velocity, and variety (Laney 2001). Yet, if data on online creative engagements across countries were available (see also Kapoor and Tagat 2020), cross-sectioned by age and gender, this would add much-needed real-time context to PISA creativity data. In sum, context exists everywhere; we hope researchers use it in future analyses wisely, especially when comparing across countries.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Endnotes

¹ It is important to clarify the difference between large-scale datasets (such as PISA) and big data using the formal definition of big data. As Laney (2001) suggests, for a dataset to be considered as big data, it must meet the 3Vs criteria: volume (large number of observations); velocity (be captured at a high frequency), and variety (contain data in various formats: images, audio etc.). PISA may not be considered part of big data, since it mainly captures quantitative data on an annual or bi-annual basis.

² Social location refers to an individual's (perceived or real) position in society based on characteristics like class, race, and socioeconomic status (Evans 2004).

³ For instance, the concept of situated cognition suggests that cognitive processes are shaped by environmental and contextual factors; we thank an anonymous reviewer for raising this point. In the PISA context, prior work has focused on the impact of the environment within which learning occurs (Coll et al. 2010). However, our use of context awareness, drawn from computing, serves a different function in the context of PISA data. Specifically, context awareness emphasizes the ability to detect, interpret, and respond to contextual information dynamically; therefore, we use context awareness as a framework for thinking about how systems (in this case statistical models, for example) can become more attuned or adapt to variations across cultural, institutional, or educational settings across nations.

⁴ This paper does not address how context is more likely to be captured in qualitative data (as opposed to quantitative datasets, such as PISA). This is important since users of PISA data (policymakers, researchers, educationists) may have far more context from qualitative data compared to quantitative data, but have less ability to compare effectively across countries (Wiseman 2010).

⁵ To check for the sensitivity of results to potential outliers among non-PISA countries, the same analysis was repeated, dropping the top four large economies in the non-PISA countries list (United States of America, India, Japan, and China). The results are similar, except that all differences show statistical significance, barring the youth employment rate (which was not statistically significant).

⁶ Popper's original 1992 quote was "Science may be described as the art of systematic oversimplification—the art of discerning what we may with advantage omit" (Stark 2018).

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